DE–2278

Sub. Code
11

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

ELECTRONIC DEVICES AND DIGITAL CIRCUITS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about EX-3 code, Gray code, BCD codes. (10)

   (b) (i) Add $(1011)_2$ and (1100). (3)

   (ii) Subtract $(1011)_2$ from $(1111)_2$. (3)

   (iii) Using 2's complement subtract 5 from 7. (4)

2. (a) Explain how transistor acts as a switch. (10)

   (b) Draw NOT, NOR, EX-OR, NAND, AND gates and give the truth table. (10)

3. (a) (i) Convert $(2246)_{10}$ into octal. (3)

   (ii) Convert $(224)_{10}$ into hexadecimal. (3)

   (iii) Convert $(0.656)_{10}$ into binary number. (4)

   (b) Prove that $AB + CD = \overline{\overline{AB} \cdot \overline{CD}}$ using De Morgan's theorem. (10)

4. (a) Draw the full subtractor circuit and explain with the truth table. (10)

   (b) Explain Quine McKluskey tabulation method for simplification of digital circuit. (10)

5. (a) Draw the RS flip flop circuit and explain its function. (10)

   (b) Explain the function of controlled shift register. (10)

6. (a) Draw the 3 bit shift register and explain how shift left takes place. (10)

   (b) Explain the function of ring counter. (10)

7. (a) Explain in detail the working function of photo diode. (10)

   (b) Explain the working function of silicon controlled rectifier. (10)

8. (a) Explain the working function of JFET with suitable diagram. (10)

   (b) Draw the OP-AMP adder circuit and explain the working function. (10)

DE–2279

Sub. Code
12

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is the "New Story" of computers and Communications? Explain. (10)

   (b) Describe the developments in the computer technology. (10)

2. (a) What is Software? Explain about various types of software. (10)

   (b) Write about Internet web browsers and specialised software. (10)

3. (a) Explain about the practical use of Communications and connectivities using telephone related services. (10)

   (b) What is Cyber ethics? Explain in detail. (10)

4. (a) Describe about compression and decompression in detail. (10)

   (b) Explain the various features of a DBMS in detail. (10)

5. (a) What is MIS? Explain its concepts in detail. (10)

   (b) Describe in detail about the six phases of system analysis and design. (10)

6. (a) Explain the various developments in the Communication technology. (10)

   (b) Write in detail about the ethics of Information Technology. (10)

7. (a) Explain about the internet and shared resources. (10)

   (b) What is Modem? Explain about Communication software in detail. (10)

8. (a) How can you organise the data in secondary storage? Explain. (10)

   (b) Explain the basic concepts of fire management systems. (10)

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DE–2280

Sub. Code
13

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

PRINCIPLES OF OPERATING SYSTEM

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the basic structure of operating system. (10)

   (b) Write about the system calls for input / output management. (10)

2. (a) Discuss briefly about process control block. (10)

   (b) What is critical section? Explain the mutual exclusion with busy waiting. (10)

3. (a) Write short notes on monitors. (10)

   (b) Describe briefly about two level scheduling. (10)

4. (a) Name the two categories of I/O devices and explain in detail. (10)

   (b) Discuss briefly about the usages of interrupt handler. (10)

5. (a) Define deadlock. Write the necessary conditions for deadlock. (6)

   (b) Explain in detail about deadlock detection and recovery. (14)

6. (a) Explain about multiprogramming with fixed partitioning. (12)

   (b) What is swapping? Explain its advantages with an example. (8)

7. (a) Write in detail about bit-map and linked-list memory management techniques. (16)

   (b) Write the difference between paging and segmentation. (4)

8. (a) Discuss about disk space management. (10)

   (b) Describe in detail about file system security and protection. (10)

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DE–2281

Sub. Code
14

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

C PROGRAMMING

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is structured programming? Explain about procedures and recursion. (12)

   (b) Write in detail about factoring methods. (8)

2. (a) Explain in detail about identifiers, constants and variables in C with example. (15)

   (b) Write short notes on precedence of operators. (5)

3. (a) Explain the following statements with examples :

   (i) Switch.

   (ii) Continue.

   (iii) Break. (15)

   (b) Write a C program to find the factorial of a given number. (5)

4. (a) What is function? Explain the categories of functions? (10)

   (b) Write a C program to store and display the student marks and result using array. (10)

5. (a) Explain in detail about evaluation of the expressions. (15)

   (b) Write a C program to find the sum of five digit number. (5)

6. (a) Explain in detail any two loop structures with example. (10)

   (b) Write a C program to find the multiplication of given two matrices. (10)

7. (a) What is pointer? Explain pointers with suitable program. (10)

   (b) What is structure? Explain about structures within structures. (10)

8. (a) Explain in detail about various file operations. (10)

   (b) Write a C program to read the employee information from a data file and display them. (10)

DE–2282

Sub. Code
15

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

ACCOUNTING FUNDAMENTALS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. Differentiate between financial accounting and management accounting.

2. Explain various accounting concepts briefly.

3. Write short notes on:
   (a) Accounting equation
   (b) Trial balance
   (c) Final accounting
   (d) Suspense accounts.

4. What do you mean by 'Accounting Ratios'? How are they useful?

5. Rectify the following errors:
   (a) Sales to Ravi Rs. 152, Posed to his account as Rs. 125
   (b) Purchased goods from Mohan Rs. 600, credited as Rs. 500
   (c) Received B/R from Banu Rs. 4,000, posted as Rs. 400

(d) Purchased furniture from Raja Rs. 505 on credit debited as Rs. 550

(e) Discount allowed Rs. 64 to Arul Credited to his account as Rs. 46

(f) Discuss received Rs. 37 from Gobu to his account as Rs. 39

(g) Purchased furniture on cash for Rs. 2,000 was not posted

(h) Purchased book is carried forward Rs. 1,000 less.

6. From the following balances of Parithi, prepare a Trading a/c, Profit and Loss a/c and Balance sheet as at 31st December 2013.

Credit Balance	Rs.
Capital	72,000
Creditors	17,440
Bills payable	5,054
Sales	1,56,364
Loan	24,000
Debit Balance:	
Debtors	7,770
Salaries	8,000
Discount	2,000
Postage	546
Bad debts	574
Interest	2,590
Insurance	834
Machinery	20,000
Stock (1.1.2013)	19,890
Purchases	1,24,184
Wages	8,600

Buildings	47,560
Furniture	32,310

Value of goods on hand on 31.12.2013 Rs. 28,600.

7. Aravind keeps his books under single entry system. Assets and Liabilities on 31.12.2012 and 31.12.2013 stood as follows:

	31.12.2012 Rs.	31.12.2013 Rs.
Cash	40	8,000
Bank balance	4,000	40,000
Stock	28,000	40,000
Sundry Debtors	60,000	80,000
Furniture	12,000	12,000
Sundry Creditors	12,000	24,000

He introduced an additional capital of Rs. 12,000 during 2013. He withdrew Rs. 28,000 for his domestic purpose. Find out the profit for 2013.

8. From the following given ratios and figures, prepare a summarized balance sheet of XYZ Co. Ltd for the year ended 31.12.2013.

(a)	Working capital	-	Rs. 60,000
(b)	Reserves and surplus	-	Rs. 40,000
(c)	Bank over draft	-	Rs. 10,000
(d)	Asset (Fixed)/ Proprietary ratio	-	0.75
(e)	Current ratio	-	2.5
(f)	Liquid ratio	-	1.5

DE–2283

Sub. Code
21

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

MICROPROCESSORS AND PERSONAL COMPUTERS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the Address bus and Control bus with neat diagram. (10)

   (b) What is an assembler? Explain. (10)

2. (a) Explain the salient features and internal architecture of 8085. (10)

   (b) What is an Interrupt? Explain. (10)

3. (a) Discuss the various addressing modes used in 8085. (10)

   (b) Name the five flags of 8085. Explain in detail. (10)

4. (a) Illustrate the method to subtract 3250 from 72532 using 1's and 2's complement. (10)

   (b) Write a program in 8085 assembly language to Convert ASCII into Hexadecimal. (10)

5. (a) With neat sketch explain the functions of EPROM. (10)

   (b) Discuss in detail about Memory Organization. (10)

6. (a) Discuss how will you interface a DMA controller with a microprocessor. (10)

   (b) Write notes on Memory mapped I/O. (10)

7. (a) Discuss in detail about the features and internal architecture of 80486. (10)

   (b) Explain about SCSI interface in detail. (10)

8. (a) Discuss in detail about the PC maintenance procedure. (10)

   (b) Explain about power protection system in detail. (10)

DE–2284

Sub. Code
22

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

OFFICE AUTOMATION

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write the importance of working with Windows. Give any two examples. (10)

   (b) State and briefly write how to create a Shortcut Icon. (10)

2. (a) Explain how to move, correct and Insert a Text with MS-WORD with example. (10)

   (b) Write brief notes on :

      (i) Printing a Document (5)

      (ii) Selecting and Copying Text. (5)

3. (a) Discuss how to create Tables with Inserting, Moving and Resizing Pictures. (10)

   (b) What is Spell Checking? Explain its importance in MS-Word. (10)

4. (a) What is a Worksheet in MS-EXCEL? Explain its applications and uses. (10)

   (b) Write brief notes on Creating and Copying formulae in MS-Excel. (10)

5. (a) Explain how to create a Database in MS-ACCESS with example. (10)

   (b) Write how to perform all operations on a table in MS-ACCESS. (10)

6. (a) Discuss the steps involved in improving the appearance of Worksheet. (10)

   (b) List and explain the various charts in MS-EXCEL. (10)

7. Explain about Creating Power point Presentation for an Institution in which you are studying. (20)

8. (a) Discuss how to create a hangout from a PowerPoint Presentation. (10)

   (b) Discuss Creating and Printing Binders. (10)

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DE–2285

Sub. Code
23

DISTANCE EDUCATION

B.C.A./ B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

SYSTEMS ANALYSIS AND DESIGN

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the elements of a system? Explain briefly. (10)

   (b) Explain the various phases involved in system development life cycle. (10)

2. (a) Enumerate the merits and demerits of questionnaire method. (10)

   (b) Discuss in detail about conducting the investigation. (10)

3. (a) Explain about technical, economical and operational feasibility study. (10)

   (b) Explain the following :

      (i) Present value. (5)

      (ii) Pay back. (5)

4. (a) Explain about tools for system analyst. (10)

   (b) Explain structured English with a suitable illustration. (10)

5. (a) Explain logical and physical system design. (10)

   (b) Explain about input design for on-line system. (10)

6. (a) Explain the types of files in file design. (10)

   (b) Explain various types of databases. (10)

7. (a) What is the need of system testing? Explain any five testing techniques and their basic objectives. (10)

   (b) List various factors which are responsible for the quality of a system. Also explain the different levels of quality assurance. (10)

8. (a) Explain all the conversion methods. Also mention an advantage and a disadvantage of each. (10)

   (b) Explain system valuation and maintenance in detail. (10)

DE–2286

Sub. Code
24

DISTANCE EDUCATION

B.C.A./B.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

OBJECT ORIENTED PROGRAMMING AND C++

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the key concepts of OPP. (10)

   (b) Explain the evolution of C++. (10)

2. (a) Describe the procedure for designing manipulators. (10)

   (b) What are single and multiple parameter manipulators? Explain. (10)

3. What is constructor? Explain any two types of constructor with examples. (20)

4. (a) What are friend functions and friend classes? (10)

   (b) Explain the features of member functions. (10)

5. (a) What are the features and uses of pointers? (10)

   (b) How can private members be accessed using pointers? Explain. (10)

6. (a) What are the rules for overloading operators? (10)

   (b) What is the difference between overloading of binary operators and unary operators? (10)

7. Explain inheritance methods with an example. (20)

8. (a) Describe the different types of polymorphism. (10)

   (b) Write a C++ program to create a class MAT of size $m \times n$ and define all possible matrix operations for MAT type object. (10)

DE–2287

Sub. Code
25

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

COMPUTER GRAPHICS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes on Application of Computer Graphics. (5)

   (b) Explain about the Input Devices and Video Display Generation in detail. (15)

2. (a) Explain Point Plotting and Line Drawing algorithms. (10)

   (b) Explain the concept of Region Filling Techniques with examples. (10)

3. (a) Explain in detail about 2-D Transformations and Principles with examples. (10)

   (b) Discuss the Composite Transformations with diagram. (10)

4. (a) Describe in detail about the 2-D Viewing and Clipping. (15)

   (b) Explain the Issues of Aspect Ratio. (5)

5. (a) Describe the 3-D Transformations Concepts in detail. (15)

(b) Explain the features of Mirror Reflection with diagram. (5)

6. (a) Describe the Cohen–Sutherland Algorithm in detail. (15)

(b) Explain the Clipping against a Finite View Volume with example. (5)

7. (a) Explain about the Matrix Representations of 2–D Transformations. (5)

(b) Explain the simple DDA Algorithm with diagram. (15)

8. (a) Describe concept of User's Model and the Command Language in detail. (10)

(b) Explain about the Styles of Command Language and Examples. (10)

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DE-2288

Sub. Code
31

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

BUSINESS COMMUNICATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write about different types of communication. (10)

   (b) Explain the following communication Goals :

   Increasing Profitability, Improving Work Quality,

   Winning Back Lost Customers, Improving Training Process. (10)

2. How are Business Letters written for placing orders and acknowledging the receipt of goods? (20)

3. (a) What are the characteristics of a good report? (10)

   (b) How are minutes recorded? (10)

4. (a) How are short speeches written? (10)

   (b) Draft a welcome address to the Inauguration of the Computer Science Association. (10)

5. Distinguish between Social Letters and Business Letters. (20)

6. (a) Write a letter from Messrs. Chand & Co., Delhi, to Messrs. Prakash Tradings, Chennai, asking for the price-list for fountain-pens, pencils, ball pens etc. (10)

(b) Draft the minutes of the First Annual General Meeting of a Public Limited Company. (10)

7. (a) What are the essentials of a Complaint Letter? (10)

(b) Write a letter from Deviprasad Company, Mumbai to Messrs. Premchand Lal and Co, Bangalore, detailing the settlement of accounts. (10)

8. (a) Draft the report of a travelling salesman on market conditions for the products of your firm. (10)

(b) Write a letter to the Editor commenting on the recent student trouble in a college. (10)

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DE–2289

Sub. Code
32

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

WINDOWS AND VISUAL BASIC

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the basic concepts of Windows programming. (10)

   (b) Explain in detail about the general format of a Window. (10)

2. (a) Discuss on the importance of control panel. (10)

   (b) Describe the various drawing tools in Windows. (10)

3. (a) Write short note on Operating modes in Windows. (10)

   (b) What is OLE? Explain. (10)

4. (a) What is Event-driven programming? Discuss. (10)

   (b) Write short note on menu system in VB. (10)

5. Briefly explain about the following controls :

   (a) Tree view control (10)

   (b) Common dialog control (10)

6. (a) How do you write save, run and test a VB code? Explain. (10)

   (b) Write a VB program to check whether a given number is prime or not. (10)

7. (a) Give the syntax and purpose for IF statement with an example. (10)

   (b) Write a VB program to find the largest among three numbers. (10)

8. (a) How do you create procedures and functions in VB? Discuss. (10)

   (b) Explain in detail about MDI Forms in VB. (10)

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DE–2290

Sub. Code
33

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

INTERNET AND JAVA PROGRAMMING

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) What are protocols? Write the importance of TCP/IP. (10)

   (b) Write short notes on TCP applications layer. (10)

2. (a) What is DNS? Explain about Mail Transfer Protocol. (10)

   (b) Discuss about Telenet. (10)

3. What are the applications of the internet? Explain them. (20)

4. (a) How does E-Commerce help the people? Explain. (10)

   (b) Explain HTTP client requests and HTTP methods. (10)

5. (a) What are the data types in java? Explain with examples. (10)

   (b) List and explain the types of operators in java. (10)

6. (a) What is an array? Give an example for multidimensional array. (10)

   (b) Write short note on type conversion and casting. (10)

7. Describe constructors and destructors with suitable examples. (20)

8. (a) Explain the different types of inheritance in java. (10)

   (b) Describe the operations on string. (10)

DE–2291

Sub. Code
34

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

RDBMS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain about schema and subschema in detail. (10)

   (b) Describe the different user interfaces. (10)

2. (a) Explain the data manipulation facilities. (10)

   (b) Explain the concept of relational model. (10)

3. (a) Explain the logical structures in oracle with example. (10)

   (b) Explain the oracle architecture. (10)

4. (a) Describe the different types of join. (10)

   (b) Describe any four embedded SQL commands. (10)

5. (a) Write about unions. (10)

   (b) Explain the data control facilities. (10)

6. Explain how to create a table, modify a table, Rename a table, copying from another table and to drop the table with examples. (20)

7. (a) Explain the client-server systems. (10)

   (b) Describe the different kinds of mapping with examples. (10)

8. (a) What is database schema? Explain. (10)

   (b) Discuss about embedded SQL precompiler. (10)

DE–2292

Sub. Code
35

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

MANAGEMENT PRINCIPLES AND TECHNIQUES

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Describe the functions and principles of management.

2. Explain the importance of communication and motivation.

3. Solve the following LPP

   Min $z = x_1 + x_2$

   Subject to the constraints

   $2x_1 + x_2 \geq 4$

   $x_1 + 7x_2 \geq 7$

   $x_1, x_2 \geq 0$.

4. Solve the following transportation problem

					Available units
	6	1	9	3	70
	11	5	2	8	55
	10	12	4	7	90
Required units	85	35	50	45	

5. The following table lists the jobs of a network along with their time estimates.

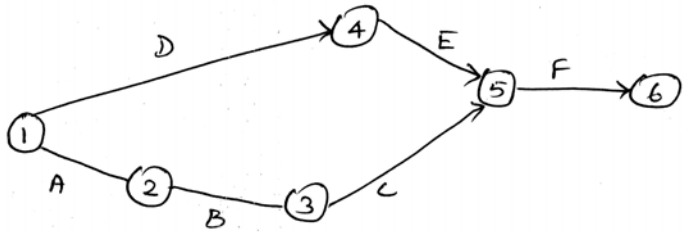
Job :	1–2	1–6	2–3	2–4	3–5	4–5	6–7	5–8	7–8
t_0 :	3	2	6	2	5	3	3	1	4
t_m :	6	5	12	5	11	6	9	4	19
t_p :	15	14	30	8	17	15	27	7	28

(a) Draw the project network

(b) Calculate the length and variance of critical path

(c) What is the probability to compute the project within 41 days?

6. In the project network shown in the figure given below the nodes are denoted by numbers and the activities by letters. The normal and crash durations of the various activities along with the associated costs are show below



Activity :	A	B	C	D	E	F
Normal duration :	8	16	14	12	15	10
Normal cost :	1800	1500	1800	2400	800	2000
Crash duration :	6	11	9	9	14	8
Crash cost :	2200	2200	2400	3000	2000	4000

Determine the least cost for 37 day schedule.

7. A Machine costs Rs. 10,000 operating costs are Rs. 500 for the first years. In the sixth and succeeding years operating costs increases by Rs. 100 per year. Assuming a 10% discount rate of money per year. Find the optimum length of time to hold the machine before we replace it.

8. A computer contains 10000 resistors. When any resistor fails, it is replaced. The cost of replacing a resisitor individually is Rs. 1 only. If all the resistors are replaced at the same time, the cost per resistor would be reduced to 35 paise. The percent survicing $s(t)$ at the end of month t is given below

t :	0	1	2	3	4	5	6
$s(t)$:	100	97	90	70	30	15	0

What is the optimum replacement plan?

DE–9028

Sub. Code
16

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2015.

LAB I — C PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Examiners has to select and give ONE question to each candidate by lot system.

Answer both (a) and (b).

1. (a) Write a C program to print the all ASCII characters.

   (b) Write a C program to accept and display your ADDRESS, using structure.

---------- Cut here ----------

2. (a) Write a C program to find simple and compound interest using FUNCTION declaration.

   (b) Write a C program to sort 10 Nos in ascending order.

---------- Cut here ----------

3. (a) Write a C program to find the factorial of a given number using FUNCTION declaration.

   (b) Write a C program to create an ADDRESS file.

4. (a) A man is paid at the hourly rate of Rs. 15 per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked per week, Calculate and print his gross weekly wage

(b) Write a C program to print the biggest of 5 Nos.

---------- Cut here ----------

5. (a) Write a C program to accept two strings. Assign a string to another variable, concatenate two strings and find the length of concatenated strings.

(b) Write a C program to find a word is PALINDROME or not.

---------- Cut here ----------

6. (a) Write a C program to compute commission earned by a salesman according to the scheme given below.

SALES AMOUNT	COMMISSION (%)
Upto Rs. 1,000	0
Above 1,000 upto 5,000	5
Above 5,000	10

(b) Write a C program to accept and display your ADDRESS using structure.

---------- Cut here ----------

7. (a) Write a C program to find simple and compound interests using FUNCTION declaration.

(b) Write a C program to accept and display your DATE OF BIRTH using structure.

8. (a) Write a C program that receives the data such as age and name of person to check the eligibility for voting. Take the condition if a person is more than 18 years old he is eligible to vote. Else display the number of years, he has to wait for voting.

(b) Write a C program to sort 5 NAMES in ascending order.

DE–9029

Sub. Code
26

DISTANCE EDUCATION

B.C.A. /B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

LAB II : C++ AND MS OFFICE

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by LOT system. Answer both (a) and (b).

1. (a) Write a C++ program to arrange the given set of numbers in ascending/descending order.

   (b) Create a table STUDENT with the following fields :

   Reg. No, name, age, sex, degree, percentage

   (i) Add 5 records using forms.

   (ii) Display the records for sex = 'M' and degree = 'BCA'

   (iii) Display the records for sex = 'F' and percentage between 80 and 90.

   (iv) Display the records for name like 'S*'.

   (v) Sort the table in ascending order of percentage.

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2. (a) Write a C++ program to read and display the students particulars using class and object.

   (b) Create the worksheet in Excel.

Year :	2000	2001	2002	2003
Sales :	20	25	35	42
Expense :	9	12	38	25

(i) Draw the graph and mention the appropriate headings

(ii) Include profit with data values and create different types of graph.

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3. (a) Write a C++ program to illustrate various arithmetic operations using a member function which are defined out of the scope of a class definition.

(b) Type a letter to BCA candidates to attend PCP classes. Format the letter and apply suitable font type, size and color options.

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4. (a) Write a C++ program to read the derived class data members from the keyboard and display the contents of the class on the screen.

(b) Using MS-Word type the title page of your office automation book.

(i) Format the title page by the following features

* Centre each line

* Choose appropriate font size, font type

(ii) Draw border lines

(iii) Use color options.

5. (a) Write a C++ program to find the square of a given number with different arguments using function overloading.

(b) Create a worksheet in excel with the following columns :

S. No , Reg. No , name, age, marks for 5 subjects, total, average, result.

(i) Fill the total column

(ii) Fill the average column

(iii) Replace the result with pass for average >= 40 else fail.

(iv) Sort first by name then age.

(v) Find the maximum and minimum for each subject.

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6. (a) Write a C++ program to create a class of objects e1 and e2. The contents of object e1 is added to the object e2 using the operator overloading technique.

(b) Create a table LIBRARY with the following fields :

Books No., Title, Author, Price, Publisher, Branch

(i) Add 5 records using forms.

(ii) Display the records for title = 'Java' and Author like 'M∗'.

(iii) Display the records for publisher = 'PHI' or 'TMH' and price between 300 and 500.

(iv) Sort the table in ascending order of book No.

DE–9030

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DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

Lab III — ORACLE AND VISUAL BASIC

(Common for all batches)

Time : Three hours Maximum : 100 marks

Examiner has to select two question ONE from ORACLE and another from VISUAL BASIC for each candidate.

The questions are given to the candidates by lot system.

VISUAL BASIC

1. Write and test a VB Program to compute and print either the SUM or PRODUCT of the first 'N' natural numbers.

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2. Write and test a VB program to compute the surface area and volume of a sphere given the radius. Use option buttons and INPUT boxes.

   Formula : A = 4 * 3.14 * R ∧ 2

   $$Y = \frac{4}{3} * 3.14 * R \wedge 3 .$$

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3. Write and test a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using label, text and option buttons.

4. Write and test a VB program that allows then user's choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

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5. Write and test a VB program to display the day of the given date and convert upper case from lower case to given word.

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ORACLE

6. Create table MARK with REGNO, NAME, TAMIL, ENGLISH, MATHS and BIOLOGY fields and perform the following.

   (a) Add 5 records.

   (b) Show data in fields REGNO, BIOLOGY and MATHS alone on the screen.

   (c) Show all fields on the screen.

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7. Create table RESULT with REGNO, NAME, MARK 1, MARK 2, MARK 3, TOTAL and RESULT fields.

   (a) Add 5 records (fill all fields except total and result fields).

   (b) Fill TOTAL field with the sum of MARK 1, MARK 2 and MARK 3.

   (c) Fill the RESULT field with 'PASS' if total ≥ 150 otherwise 'FAIL'.

8. Create table BIODATA with NAME, AGE, SEX, DEGREE and CITY fields

   (a) Display all records of MALE employees.

   (b) Delete all B.Sc. degree holders and then display the table contents.

   (c) Display all records having AGE<20 and CITY is TRICHY.

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9. Create table SALES with SALE NO, SALE NAME, SALE AMT, COMM and SALEDATE fields.

   (a) Display all the records.

   (b) Display the records having commission <1000.

   (c) Display the average sales and average commission.

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10. Create table NUMB with NUM, ABSNO, SQRTNO, ROUNDNO, TRUNCNO, SQRNO, CUBNO fields.

   (a) Replace ABSNO with absolute value of give numbers.

   (b) Replace SQRNO with square of given number.

   (c) Display all the records.